

AKASH TIWARI

RED TEAM & BLUE TEAM CYBERSECURITY ENGINEER | AI & CS SCHOLAR

Email: akashkumartiwariofficial@gmail.com Location: Patna, Bihar, India Focus: Offensive & Defensive Security, AI Threat Detection

PROFESSIONAL SUMMARY

Computer Science scholar at IIT Patna specializing in dual-spectrum cybersecurity and Artificial Intelligence. Hands-on experience in offensive Red Team methodologies (Vulnerability Exploitation, Penetration Testing, OSINT, Payload Development) and defensive Blue Team architectures (Incident Response, Network Hardening, Cloud Security, Threat Mitigation). Advanced capability in high-performance security tooling development using **Rust** and **Python**.

EDUCATION

Indian Institute of Technology, Patna Jan 2026 – Dec 2030

Bachelor of Science & Master of Science in Computer Science (Specialization in AI & Cybersecurity)

Indian Institute of Technology, Guwahati Oct 2024 – Oct 2026

Advanced Program in Cybersecurity

TECHNICAL EXPERTISE

Core Languages:	Python, Rust, C/C++, Shell Scripting
Red Team (Offensive):	Penetration Testing, Kali Linux, Nmap, Burp Suite, OSINT, Cryptographic Analysis, Reverse Engineering
Blue Team (Defensive):	Incident Response, Cloud Security, Network Infrastructure Hardening, Threat Detection, SIEM Fundamentals
AI & Cryptography:	Voice Cloning Fraud Detection Models, Applied Cryptography, Machine Learning for Threat Intelligence

CYBERSECURITY DOMAIN & PROJECT EXPERIENCE

Offensive Security (Red Team)

- Vulnerability Assessment & Web App Exploitation:** Performed structural network reconnaissance and vulnerability assessments using **Kali Linux**, **Nmap**, and **Burp Suite** to map attack surfaces and isolate security flaws.
- Offensive Security Tooling (Rust & Python):** Engineered lightweight, custom enumeration and threat-simulation scripts in **Rust** for fast, reliable system probing and cryptographic verification.
- OSINT & Reconnaissance Operations:** Conducted thorough OSINT investigations to uncover credential leaks, misconfigurations, and external surface exposures.

Defensive Security & AI Defense (Blue Team)

- Incident Response & Traffic Monitoring:** Analyzed attack vectors and network logs to construct effective mitigation plans, ensuring zero-trust protection against malicious intrusions.
- AI-Powered Voice Fraud Detection System:** Developed research architecture focused on real-time detection of AI-assisted voice cloning fraud targeting financial systems and credit card voice authentication.
- Cloud & Infrastructure Defense:** Hardened cloud configurations and network perimeters against exfiltration vectors, applying defense-in-depth security principles.

PUBLICATIONS & LEADERSHIP

- Published Author:** Authored and published the book *"The Civic Sense of Indian People"* (Published by BookOf Publishers).